LAWS OF EXPONENTS

Each law with its restriction

Five of the eight laws come with a condition; the condition is part of the law.

✓ NO RESTRICTION ◐ RESTRICTED

Product ✓ NO RESTRICTION

$$a^m \cdot a^n = a^{m+n}$$

the bases must match

$x^3 \cdot x^5 = x^8$

Quotient ◐ RESTRICTED

$$\frac{a^m}{a^n} = a^{m-n}$$

$a \neq 0$

$\frac{x^7}{x^2} = x^5$

Power of a power ✓ NO RESTRICTION

$$(a^m)^n = a^{mn}$$

not the stacked a^{m^n}

$(x^3)^4 = x^{12}$

Power of a product ✓ NO RESTRICTION

$$(ab)^n = a^n b^n$$

for products only, never sums

$(3x)^4 = 81x^4$

Power of a quotient ◐ RESTRICTED

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$b \neq 0$

$\left(\frac{2}{5}\right)^3 = \frac{8}{125}$

Negative exponent ◐ RESTRICTED

$$a^{-n} = \frac{1}{a^n}$$

$a \neq 0$

$4^{-2} = \frac{1}{16}$

Zero exponent ◐ RESTRICTED

$$a^0 = 1$$

$a \neq 0$; 0^0 is excluded

$7^0 = 1$ and $(-3)^0 = 1$

Rational exponent ◐ RESTRICTED

$$a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$$

$a \geq 0$ when n is even

$8^{2/3} = (\sqrt[3]{8})^2 = 4$